

**A. Raw Test Results — 10 Trials**

| <b>Trial</b> | <b>Distance (cm)</b> | <b>Detection (s)</b> | <b>Suppression (s)</b> | <b>Alert (s)</b> | <b>Outcome</b> |
|--------------|----------------------|----------------------|------------------------|------------------|----------------|
| 1            | 50                   | 1.3                  | 5.2                    | 6.1              | Success        |
| 2            | 75                   | 1.5                  | 5.6                    | 5.9              | Success        |
| 3            | 100                  | 1.4                  | 5.5                    | 6.3              | Success        |
| 4            | 150                  | 1.6                  | 5.9                    | 6.5              | Success        |

|    |     |     |     |     |                                 |
|----|-----|-----|-----|-----|---------------------------------|
| 5  | 200 | 1.8 | 6.1 | 6.4 | Success                         |
| 6  | 75  | 1.2 | 5.3 | 6.0 | Success                         |
| 7  | 100 | 1.5 | 5.5 | 6.2 | Success                         |
| 8  | 250 | 2.2 | 6.5 | 6.8 | <b>Partial —<br/>over-shoot</b> |
| 9  | 100 | 1.4 | 5.4 | 6.0 | Success                         |
| 10 | 50  | 1.3 | 5.2 | 5.9 | Success                         |

**Mean detection latency: 1.52s | Mean suppression time: 5.62s | Mean alert delivery: 6.21s | 9/10 full success**

## B. Risk Register

| #  | Risk                             | Likelihood | Impact | Mitigation                                                    |
|----|----------------------------------|------------|--------|---------------------------------------------------------------|
| R1 | Indoor GPS drift                 | High       | Medium | Antenna repositioning; defer to last-known indoor reference   |
| R2 | IR sensor saturation in sunlight | Medium     | High   | Threshold calibration + AND-logic with DHT11                  |
| R3 | Water damage to electronics      | Medium     | High   | Servo-aimed directional spray; water-proofing of MCU          |
| R4 | Battery exhaustion mid-mission   | Medium     | High   | Low-voltage cutoff; idle sleep mode                           |
| R5 | Wi-Fi unavailable during fire    | Medium     | Medium | Local SD-card log + retry; fallback SMS gateway (future work) |
| R6 | Mechanical jam in chassis        | Low        | High   | Pre-mission self-test of motor encoders                       |